

Exact equation discovery from implicit, rational, and polynomial data

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Equation discovery aims to recover mathematical relationships directly from data, making it a promising tool for scientific research across many fields, from physics and chemistry, to molecular biology and biotechnology, materials science, and environmental sciences, including isotope tracing. While most current methods aim to recover approximate equations from noisy measurements, we focus on exact equation discovery, where the objective is to identify equations that describe noise-free data perfectly.

Integer sequences provide a natural benchmark for this task. For instance, from the Fibonacci sequence (0, 1, 1, 2, 3, 5, 8, . . .), one can ask whether an algorithm can recover the recurrence ($a_n = a_{n-1} + a_{n-2}$). In earlier work, we evaluated methods based on Diophantine equation solving and Gröbner basis computation on explicit equations arising from OEIS sequences, including some non-linear examples. However, only linear equations were studied at scale. In this study, we extend that setting to more general classes of equations.

We constructed a benchmark of datasets generated from 20 implicit equations, 129 rational equations, and 2,257 polynomial equations, and used it to compare our methods with state-of-the-art approaches. The results show a clear pattern: Gröbner basis methods remain the only viable option for implicit and rational equations, whereas polynomial equations are handled significantly better by Diophantine-equation-based methods. The main drawback of the latter is high computational complexity. These findings show that exact equation discovery benefits from combining ideas from machine learning, symbolic computation, and algebra, and that different equation classes require different computational tools.

Related publications:

[1] B. Gec, S. Džeroski, and L. Todorovski, *Discovery of exact equations for integer sequences*, Mathematics, vol. 12, no. 23, 2024, issn: 2227-7390. doi: 10.3390/math12233745.

[2] B. Gec, S. Džeroski, and L. Todorovski, “Discovery of exact equations via computing the Gröbner basis”, under minor revision at Machine Learning, ISSN: 1573-0565, Springer Nature.

Other equation discovery publications:

[3] B. Gec, N. Omejc, J. Brence, S. Džeroski, and L. Todorovski, “Discovery of differential equations using probabilistic grammars”, in *Discovery Science*, P. Pascal and D. Ienco, Eds., Cham: Springer Nature Switzerland, 2022, pp. 22-31, isbn: 978-3-031-18840-4.

[4] N. Omejc, B. Gec, J. Brence, L. Todorovski, and S. Džeroski, “Probabilistic grammars for modeling dynamical systems from coarse, noisy, and partial data”, *Machine Learning*, vol. 113, no. 10, pp. 7689-7721, Oct. 2024, issn: 1573-0565. doi: 10.1007/s10994-024-06522-1.